

COLLISION-3M Scaling Experiment Results

This document presents the results of the COLLISION-3M scaling experiment, comparing it directly against the baseline COLLISION-1.46M model. Both models were trained on the identical dataset (collision_dataset_v4) under identical hyperparameter conditions on CPU for exactly 1,536,000 training tokens (1,500 steps).

Direct Comparison Metrics

Metric	COLLISION-1.46M (Base)	COLLISION-3M (Exp)	Change (%)
Model Parameters	1,462,464	3,375,680	+130.82%
Best Val Loss	1.9363	0.9663	-50.10%
Best Val Perplexity	6.93	2.63	-62.08%
Total Train Time (s)	490.5s	1067.4s	+117.62%
Avg Inference (tok/s)	200.83	92.12	-54.13%

parameter capacity scaling (+130.8%) yielded a significant 50.1% decrease in Validation Loss and a 62.1% decrease in Perplexity under identical training

Generation Benchmark Outputs (COLLISION-3M)

Prompt: "What is artificial intelligence?" (82.66 tok/s | 50 tokens)

What is artificial intelligence? this and intelligence sequence and and maximize necessary intelligence next maximize loss. to and data domain.

Transformers self-attention adapts hidden maintaining

Prompt: "The future of technology" (98.91 tok/s | 50 tokens)

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COLLISION-3M Loss Curve

